Solid State Physics — Week 1 Assignment Questions & Answers

Question 1

Transfer of electrons takes place in

(a) Ionic Bonding

(b) Covalent Bonding

(c) Metallic Bonding

(d) None of these

Accepted answer: (a)

Question 2

Among the following bonds which one is the weakest?

(a) Covalent Bond

(b) Ionic Bond

(c) Metallic Bond

(d) Van der Waal's bond

Accepted answer: (d)

Question 3

Van der Waal's bond is formed due to

(a) Electron and ion cores interaction

(b) Dipole-Dipole interaction

(c) Coulomb interaction

(d) Sharing of electrons

Accepted answer: (b)

Question 4

What type of bonding exists between water molecules in ice?

(a) Covalent Bond

(b) Ionic Bond

(c) Hydrogen bond

(d) Metallic bond

Accepted answer: (c)

Question 5

Which of the following bond is directional?

(a) Covalent Bond

(b) Ionic Bond

(c) Metallic bond

(d) Van der Waal's bond

Accepted answer: (a)

Question 6

Which of the following has hydrogen bonding?

(a) CH_4

(b) CsCl

(c) NaCl

(d) HF

Accepted answer: (d)

Question 7

The coulomb interaction energy between two adjacent ions in a solid is given by $U = \left(-\frac{A}{r}+\frac{B}{r^n}\right)$ esu units. Find out the bond length of the system at equilibrium ion separation.

(a) $\left(\frac{nB}{A}\right)^{\frac{1}{n}-1}$

(b) $\left(\frac{nB}{A}\right)^{\frac{1}{n-1}}$

(c) $\left(\frac{B}{A}\right)^{n+1}$

(d) $\left(\frac{B}{A}\right)^{\frac{1}{n}-1}$

Accepted answer: (b)

Question 8

In Kcl crystal the energy of interaction between adjacent ions is given by $U = \left(-\frac{1.436}{r} + \frac{5.86\times10^{-6}}{r^9}\right)$ energy units. Determine the equilibrium energy at equilibrium ion separation.

(a) -3.31 energy units

(b) -4.57 energy units

(c) 5.21 energy units

(d) 6.67 energy units

Accepted answer: (b)

Question 9

Calculate the cohesive energy of copper (FCC) given the bond energy between two copper atoms is 56.4 KJ/mol. Consider effective number of bonds in FCC is 6 and cohesive energy to be energy per 6 bonds.

(a) 256.4 KJ/mol

(b) 338.4 KJ/mol

(c) 676.8 KJ/mol

(d) 225.6 KJ/mol

Accepted answer: (b)

Question 10

The potential energy of a system of two atoms is given by $U = \left(-\frac{A}{r^2} + \frac{B}{r^{10}}\right)$. A stable molecule is formed with release of -8 eV of energy when the interatomic distance is 2.8 Å. Calculate the values of A and B.

(a) $1.25 \times 10^{-37}\, J - m^2$, $9.44 \times 10^{-115}\, J - m^{10}$

(b) $2.15 \times 10^{-35}\, J - m^2$, $6.25 \times 10^{-120}\, J - m^{10}$

(c) $3.44 \times 10^{-30}\, J - m^2$, $5.15 \times 10^{-125}\, J - m^{10}$

(d) $4.14 \times 10^{-25}\, J - m^2$, $4.67 \times 10^{-130}\, J - m^{10}$

Accepted answer: (a)